QM II Homework Assignment 3

Tensor Operators and the Wigner-Eckart Theorem

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1. Vector Operators

a. Show that the momentum operator, $\vec{P}$ is a vector operator by showing that its components satisfy $[P_i,\, J_j] = i\hbar\epsilon_{ijk}P_k$.

b. The matrix for rotation about the $\hat{z}$ axis is

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Compute the rotated momentum vector with this matrix.

c. Transform the vector operator into a spherical tensor of rank $k = 1$ using (15.3.5). Rotate the transformed spherical tensor operator using $D^{(1)}$ from problem 1.b on homework 2. Transform the rotated spherical tensor back into a vector operator and compare your result to pare (b) above.

Part a

Lets expand the j^{th} component of the angular momentum operator in tensor notation.

$$J_i = (R \times P)_i = \epsilon_{jlm}R_lP_m$$

We can use this to expand the commutator in question.

$$\begin{aligned} [P_i,\, J_j] &= P_iJ_j - J_jP_i \\ &= P_i\epsilon_{jlm}R_lP_m - \epsilon_{jlm}R_lP_mP_i \\ &= \epsilon_{jlm}\left(P_iR_lP_m - R_lP_mP_i\right) \end{aligned}$$

Recall the commutator of position and momentum.

$$[R_i,\, P_j] = i\hbar\delta_{ij} \quad \Rightarrow \quad P_jR_i = R_iP_j - i\hbar\delta ij$$

Hence,

$$\begin{aligned} [P_i,\, J_j] &= \epsilon_{jlm}\left[(R_lP_i - i\hbar\delta_{il})P_m - R_lP_m\right] \\ &= \epsilon_{jlm}(-i\hbar\delta_{il}P_m) \\ &= -i\hbar\epsilon_{jim}P_m \\ &= i\hbar\epsilon_{ijm}P_m \end{aligned}$$

which proves that the momentum is a vector operator.

Part b

The rotated vector is given simply by:

$$\vec{P}' = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix} = \begin{pmatrix} P_x\cos\theta - P_y\sin\theta \\ P_x\sin\theta + P_y\cos\theta \\ P_z \end{pmatrix}$$

Part c

To transform this into a spherical tensor of rank 1, we simply combine the components in the following way.

$$T_1 = \begin{pmatrix} \frac{1}{\sqrt{2}}(P_x - iP_y) \\ P_z \\ \frac{-1}{\sqrt{2}}(P_x + iP_y) \end{pmatrix}$$

We can now rotate this about the $\hat{z}$ axis by acting on it with $D^(1)_z$.

$$T_1' = \begin{pmatrix} e^{-i\theta} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\theta} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}(P_x - iP_y) \\ P_z \\ \frac{-1}{\sqrt{2}}(P_x + iP_y) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}}\left[(P_x\cos\theta - P_y\sin\theta) - i(P_x\sin\theta + P_y\cos\theta)\right] \\ P_z \\ \frac{-1}{\sqrt{2}}\left[(P_x\cos\theta - P_y\sin\theta) + i(P_x\sin\theta + P_y\cos\theta)\right] \end{pmatrix}$$

By comparing this with the spherical tensor transformation we can easily recognize the components of P'.

$$P' = \begin{pmatrix} P_x\cos\theta - P_y\sin\theta \\ P_x\sin\theta + P_y\cos\theta \\ P_z \end{pmatrix}$$

Which is exactly what we found in part b.

2. k = 2 Spherical Tensor Operators

- **a.** Express the five $l = 2$ spherical harmonics in Cartesian coordinates.
- **b.** Since multiplication by spherical harmonics is an example of a spherical tensor operator, obtain a spherical tensor operator, T_2^q, from linear combination of X^2, Y^2, Z^2, XY, YZ, XZ.
- **c.** Verify that this T_2^2 satisfies the commutation relations (15.3.11) (Do this only for the commutation relations with J_z and J_-. The commutator with J_+ is similar to J_-.)

Part a

We begin by recalling the following conversions from Cartesian to spherical coordinates.

$$x = r\cos\phi\sin\theta \quad y = r\sin\phi\sin\theta \quad z = r\cos\theta$$

Which lead to

$$\begin{aligned} Y_2^2 &= \sqrt{\frac{15}{32\pi}} e^{2i\phi}\sin^2\theta \\ &= \sqrt{\frac{15}{32\pi}} \left(\cos^2\phi + 2i\sin\phi\cos\phi - \sin^2\phi\right)\sin^2\theta \\ &= \sqrt{\frac{15}{32\pi}} \frac{(x+iy)^2}{r^2} \end{aligned}$$

$$\begin{aligned} Y_2^{-2} &= \sqrt{\frac{15}{32\pi}} e^{-2i\phi}\sin^2\theta \\ &= \sqrt{\frac{15}{32\pi}} \frac{(x-iy)^2}{r^2} \end{aligned}$$

$$\begin{aligned} Y_2^1 &= -\sqrt{\frac{15}{8\pi}} e^{i\phi} \cos\theta \sin\theta \\ &= -\sqrt{\frac{15}{8\pi}} (\cos\phi + i\sin\phi) \sin\theta \sin\theta \\ &= -\sqrt{\frac{15}{8\pi}} \frac{z(x+iy)}{r^2} \end{aligned}$$

$$\begin{aligned} Y_2^{-1} &= \sqrt{\frac{15}{8\pi}} e^{-i\phi} \cos\theta \sin\theta \\ &= \sqrt{\frac{15}{8\pi}} \frac{z(x-iy)}{r^2} \end{aligned}$$

$$\begin{aligned} Y_2^0 &= \sqrt{\frac{5}{16\pi}} (3\cos^2\theta - 1) \\ &= \sqrt{\frac{5}{16\pi}} \left(\frac{3z^2}{r^2} - 1\right) \end{aligned}$$

Part b

Since multiplying by an overall radial function does not affect the angular component, we can factor out the constants out front and the $\frac{1}{r^2}$ dependence common to all of the spherical harmonics. This will leave the eigenvectors and eigenvalues of the tensor operator unaffected.

$$T_2 = \begin{pmatrix} (x-iy)^2 \\ 2z(x-iy) \\ \sqrt{\frac{2}{3}}(2z^2 - x^2 - y^2) \\ -2z(x+iy) \\ (x+iy)^2 \end{pmatrix}$$

Part c

We will begin by working out the following commutation relation.

$$\begin{aligned} [x_i x_j,\, J_k] &= x_i[x_j,\, J_k] + [x_i,\, J_k]x_j \\ &= x_i i\hbar\epsilon_{jkl}x_l + i\hbar\epsilon_{ikl}x_l x_j \\ &= i\hbar(x_i x_l \epsilon_{jkl} + x_j x_l \epsilon_{ikl}) \end{aligned}$$

We are now in position to compute the necessary commutators.

$$\begin{aligned} [J_z,\, T_2^2] &= [J_z,\, (x^2 + 2ixy - y^2)] \\ &= 2i\hbar xy - 2i(i\hbar(x^2 - y^2)) + 2i\hbar xy \\ &= 2\hbar T_2^2 \end{aligned}$$

$$\begin{aligned} [J_-, T_2^2] &= [J_x - iJ_y,\, x^2 + 2ixy - y^2] \\ &= [J_x,\, x^2 + 2ixy - y^2] - i[J_y,\, x^2 + 2ixy - y^2] \\ &= -2\hbar xy - 2i\hbar yz - 2\hbar xz - 2i\hbar yz \\ &= -4\hbar(z(x+iy)) \\ &= -2\hbar T_2^1 \end{aligned}$$

3. Wigner-Eckart Theorem: In computing optical transition rates, we will need to evaluate matrix elements like $\langle \alpha_2 j_2 m_2 | \hat{\epsilon} \cdot \vec{R} | \alpha_1 j_1 m_1 \rangle$, where $\hat{\epsilon}$ is in the direction of the polarization of the electric field. Compute the angular dependence (dependence on $\hat{\epsilon}$ of this matrix element) by expressing $\vec{R}$ as a sum of spherical tensor operators and using the Wigner-Eckart theorem.

If we express the position vector in terms of its components

$$\vec{R} = R_x\hat{x} + R_y\hat{y} + R_z\hat{z}$$

then we can express it in terms of the spherical tensor operator states.

$$\vec{R} = \frac{1}{\sqrt{2}}(T_1^1 + T_1^{-1})\hat{x} + -\frac{i}{\sqrt{2}}(T_1^1 - T_1^{-1})\hat{y} + T_1^0\hat{z}$$

Lets write out $\hat{\epsilon}$ explicitly.

$$\hat{\epsilon} = \cos\phi\sin\theta\hat{x} + \sin\phi\sin\theta\hat{y} + \cos\theta\hat{z}$$

From which we can read off $\hat{\epsilon}\cdot\vec{R}$.

$$\begin{aligned}\hat{\epsilon}\cdot\vec{R} &= \frac{1}{\sqrt{2}}\cos\phi\sin\theta(T_1^1 + T_1^{-1}) - \frac{i}{\sqrt{2}}\sin\phi\sin\theta(T_1^1 - T_1^{-1}) + \cos\phi T_1^0 \\ &= \frac{1}{\sqrt{2}}e^{-i\phi}\sin\theta T_1^1 + \frac{1}{\sqrt{2}}e^{i\phi}\sin\theta T_1^{-1} + \cos\theta T_1^0\end{aligned}$$

Hence,

$$\begin{aligned}\langle\alpha_2 j_2 m_2|\hat{\epsilon}\cdot\vec{R}|\alpha_1 j_1 m_1\rangle &= \frac{1}{\sqrt{2}}e^{-i\phi}\sin\theta\langle\alpha_2 j_2 m_2|T_1^1|\alpha_1 j_1 m_1\rangle \\ &+ \frac{1}{\sqrt{2}}e^{i\phi}\sin\theta\langle\alpha_2 j_2 m_2|T_1^{-1}|\alpha_1 j_1 m_1\rangle \\ &+ \cos\theta\langle\alpha_2 j_2 m_2|T_1^0|\alpha_1 j_1 m_1\rangle \\ &= \langle\alpha_2 j_2 m_2||T_1||\alpha_1 j_1 m_1\rangle\left(\frac{1}{\sqrt{2}}e^{-i\phi}\sin\theta\langle j_2 m_2|11,\, j_1 m_1\rangle\right. \\ &\left.+\frac{1}{\sqrt{2}}e^{i\phi}\sin\theta\langle j_2 m_2|1-1,\, j_1 m_1\rangle + \cos\theta\langle j_2 m_2|10,\, j_1 m_1\rangle\right)\end{aligned}$$

So, we can see from the selection rules that the first requirement is that $j_1 + 1 \geq j_2 \geq |j_1 - 1|$. If this requirement is fulfilled, then the angular dependence depends on the m selection rules.

$$\begin{aligned} m_2 = m_1 - 1 &\quad \sin\theta e^{i\theta} \\ m_2 = m_1 + 1 &\quad \sin\theta e^{-i\theta} \\ m_2 = m_1 &\quad \cos\theta \end{aligned}$$

Bonus: Shankar Exercise 15.3.4

(a) Using $\langle jj|jj,\, 10\rangle = [j(j+1)]^{1/2}$ show that

$$\langle\alpha j||J_1||\alpha' j'\rangle = \delta_{\alpha\alpha'}\delta jj'\hbar[j(j+1)]^{1/2}$$

(b) Using $\mathbf{J}\cdot\mathbf{A} = J_z A_z + \frac{1}{2}(J_- A_+ + J_+ A_-)$ (where $A_\pm = A_x \pm iA_y$) argue that

$$\langle\alpha' jm'|\mathbf{J}\cdot\mathbf{A}|\alpha jm\rangle = c\langle\alpha' j||A||\alpha j\rangle$$

where c is a constant independent of α, α' and $\mathbf{A}$. Show that $c = \hbar[j(j+1)]^{1/2}\delta_{mm'}$.

(c) Using the above, show that

$$\langle\alpha' jm'|A^q|\alpha jm\rangle = \frac{\langle\alpha' jm|\mathbf{J}\cdot\mathbf{A}|\alpha jm\rangle}{\hbar^2 j(j+1)}\langle jm'|J^q|jm\rangle$$

Part a

From the Wigner-Eckart theorem, we can see that

$$\langle \alpha_2 j_2 || J_1 || \alpha_1 j_1 \rangle = \frac{\langle \alpha_2 j_2 m_2 | J_1^0 | \alpha_1 j_1 m_1 \rangle}{\langle j_2 m_2 | kq,\, j_1 m_1 \rangle}$$

Since $J_1^0 = J_z$ and since $j = m$ in this case, the numerator is easily seen to be

$$\langle \alpha_2 j_2 || J_1 || \alpha_1 j_1 \rangle = \frac{m_1 \hbar \delta_{\alpha_1 \alpha_2} \delta_{j_1 j_2} \delta_{m_1 m_2}}{\langle j_2 m_2 | kq,\, j_1 m_1 \rangle} = \hbar \delta_{\alpha_1 \alpha_2} \delta_{j_1 j_2} \sqrt{j(j+1)}$$

Part b

We can expand $\mathbf{J} \cdot \mathbf{A}$ to obtain

$$\mathbf{J} \cdot \mathbf{A} = J_z A_0^1 + \frac{1}{\sqrt{2}} (J_- A_1^1 - J_+ A_1^{-1})$$

Hence, we can expand the desired matrix elements by using the Wigner-Eckart theorem.

$$\begin{aligned} \langle \alpha' j m' | \mathbf{J} \cdot \mathbf{A} | \alpha j m \rangle = \langle \alpha' j || A_1 || \alpha j \rangle \, \Big(& m' \hbar \langle j m' | j m,\, 10 \rangle \\ & - \frac{1}{\sqrt{2}} \hbar \sqrt{(j - m')(j + m' + 1)} \langle j (m' + 1) | j m,\, 11 \rangle \\ & + \frac{1}{\sqrt{2}} \hbar \sqrt{(j + m')(j - m' + 1)} \langle j (m' - 1) | j m,\, 11 \rangle \Big) \end{aligned}$$

Hence,

$$\langle \alpha' j m' | \mathbf{J} \cdot \mathbf{A} | \alpha j m \rangle = c \langle \alpha' j || A || \alpha j \rangle$$

In order to determine c consider

$$\langle j m' | j m,\, lq \rangle = \frac{\langle j m' | J_1^q | j m \rangle}{\hbar \sqrt{j(j+1)}} \tag{1}$$

Substituting this into the portion corresponding to c above yields

$$c = \hbar \sqrt{j(j+1)} \delta_{m,m'}$$

Part c

From the Wigner-Eckart theorem, we see that

$$\langle \alpha' j m' | A_1^q | \alpha j m \rangle = \langle \alpha' j || A_1 || \alpha j \rangle \langle j m' | j m,\, lq \rangle$$

Using the results of part b this becomes

$$\langle \alpha' j m' | A_1^q | \alpha j m \rangle = \frac{\langle \alpha' j m' | \mathbf{J} \cdot \mathbf{A} | \alpha j m \rangle}{\hbar \sqrt{j(j+1)}} \langle j m' | j m,\, lq \rangle$$

Finally, using equation 1 this becomes

$$\langle \alpha' j m' | A_1^q | \alpha j m \rangle = \frac{\langle \alpha' j m' | \mathbf{J} \cdot \mathbf{A} | \alpha j m \rangle}{\hbar^2 j(j+1)} \langle j m' | J^q | j m \rangle$$